$c_i = d_i$ for $1 \le i \le p$, which shows that the two representations in (1) are actually the same.

Definition: Suppose the set $\mathcal{B} = \{b_1, \dots, b_p\}$ is a basis for a subspace H. For each x in H, the **coordinates of x relative to the basis $\mathcal{B}$** are the weights $c_1, \dots, c_p$ such that $x = c_1 b_1 + \dots + c_p b_p$, and the vector in $\mathbb{R}^p$

$$[x]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$$

is called the **coordinate vector of x (relative to $\mathcal{B}$)** or the **$\mathcal{B}$-coordinate vector of x.** □

Example 1: Let $v_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}$, $v_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$, $x = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$, and $\mathcal{B} = \{v_1, v_2\}$. Then $\mathcal{B}$ is a basis for $H = \text{Span}\{v_1, v_2\}$ because v_1 and v_2 are linearly independent. Determine if x is in H, and if it is, find out the coordinate vector of x relative to $\mathcal{B}$.

Solution: By Defn. of Span, $x \in H$ iff $x = c_1 v_1 + c_2 v_2$ for some $c_1, c_2 \in \mathbb{R}$. By Defn. of Ax, $x = [v_1\ v_2]\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$. By Theorem 3 from Chapter 1, this has the same solution set as the system of equations corresponding to the augmented matrix $[v_1\ v_2\ x]$.

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \xrightarrow[\times 1/3]{\text{Row1} = \text{Row1}} \begin{bmatrix} 1 & -1/3 & 1 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \xrightarrow[\times 1/6]{\text{Row2} = \text{Row2}} \begin{bmatrix} 1 & -1/3 & 1 \\ 1 & 0 & 2 \\ 2 & 1 & 7 \end{bmatrix}$$

$$\xrightarrow[-\text{Row1}]{\text{Row2} = \text{Row2}} \begin{bmatrix} 1 & -1/3 & 1 \\ 0 & 1/3 & 1 \\ 2 & 1 & 7 \end{bmatrix} \xrightarrow[-2\times\text{Row1}]{\text{Row3} = \text{Row3}} \begin{bmatrix} 1 & -1/3 & 1 \\ 0 & 1/3 & 1 \\ 0 & 5/3 & 5 \end{bmatrix} \xrightarrow[-5\times\text{Row2}]{\text{Row3} = \text{Row3}}$$

$$\begin{bmatrix} 1 & -1/3 & 1 \\ 0 & 1/3 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\times 3]{\text{Row2} = \text{Row2}} \begin{bmatrix} 1 & -1/3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

The corresponding system of equations:

1. $c_1 - \frac{c_2}{3} = 1 \rightarrow c_1 - \frac{3}{3} = 1 \Rightarrow c_1 = 2.$
2. $c_2 = 3$

By Defn, the coordinate vector of x relative to H: $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

□